Interim Report - Change Point Analysis of Brent Oil Prices

Birhan Energies | Task 1 Deliverable | 13 Jul 2026

1. Business Context

Birhan Energies is analysing how major political and economic events - conflicts in oil-producing regions, international sanctions, and OPEC policy decisions - have shaped Brent crude oil prices, to support investment, policy, and operational decisions. This report covers Task 1: the analysis workflow, a structured events dataset, and initial exploratory findings on the price series.

2. Planned Analysis Steps (summary)

1. Load and clean the daily Brent price series (20-May-1987 to 14-Nov-2022).

2. Exploratory analysis: trend, stationarity, volatility (this report).

3. Compile a structured dataset of major events (data/events.csv, 17 events).

4. Build a Bayesian change point model in PyMC (discrete-uniform prior on switch point *tau*, before/after means, pm.math.switch, Normal likelihood); run MCMC via pm.sample().

5. Check convergence (r_hat, trace plots), identify the change point from the posterior of *tau*, and quantify the shift in price level.

6. Associate detected change points with researched events and report quantified, hypothesis-framed impact statements.

7. Build a Flask + React dashboard exposing prices, change points, and event correlations (Task 3).

Full detail: *docs/analysis_workflow.md*. Assumptions, limitations, and the correlation-vs-causation discussion: *docs/assumptions_and_limitations.md*.

3. Events Dataset

data/events.csv contains 17 major events (1990-2022) spanning geopolitical conflicts, OPEC policy decisions, sanctions, and economic shocks, each with an approximate start date, category, description, and expected price direction - e.g. Iraq's invasion of Kuwait (1990-08-02), the 2014 OPEC decision not to cut output, the 2020 Saudi-Russia price war, and Russia's invasion of Ukraine (2022-02-24). This list will be cross-referenced against detected change points in Task 2.

4. Initial EDA Findings

Dataset: 9,011 daily observations, 20-May-1987 to 14-Nov-2022. Price range $9.10-$143.95/barrel (mean $48.42, std $32.86).

4.1 Trend

The raw series (Fig. 1) shows multiple distinct regimes rather than one stable trend: a calm period below ~$25/bbl through the late 1990s; a sustained uptrend from 2002 peaking near $147/bbl in mid-2008; the 2008 financial-crisis crash; a 2011-2014 plateau around $105-125/bbl; the late-2014 OPEC-driven collapse; a COVID-19 crash to single digits in April 2020; and the 2022 spike following Russia's invasion of Ukraine.

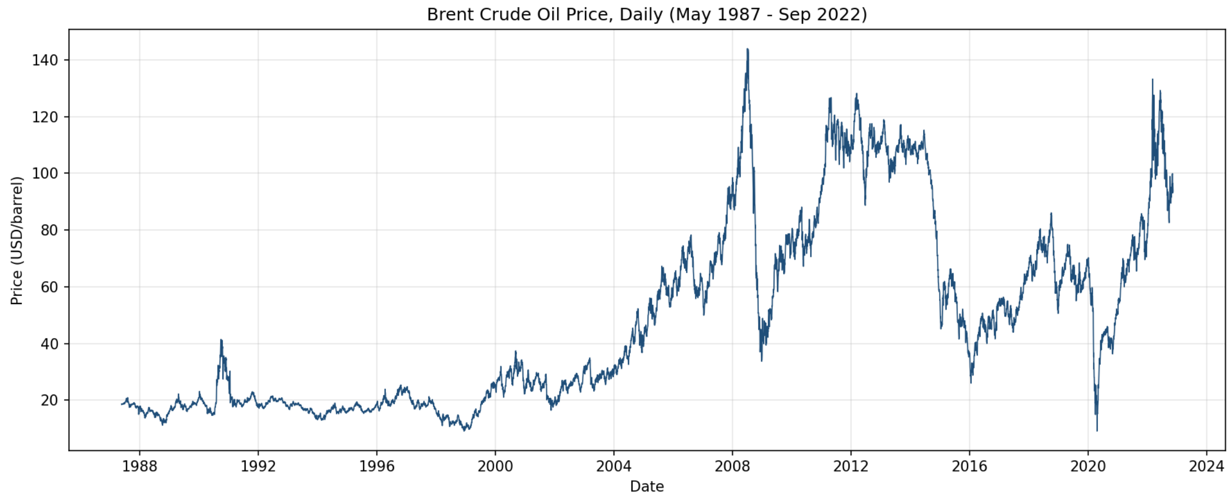

Figure 1. Brent crude oil daily price, 1987-2022. Vertical structure visibly aligns with major shocks (1990-91 Gulf War, 2008 financial crisis, 2014 OPEC glut, 2020 COVID-19, 2022 Ukraine invasion).

4.2 Stationarity

Series	ADF p-value	KPSS p-value	Conclusion
Price level	0.289	0.010	Non-stationary
Log return	~ 0.000	0.100	Stationary

Price levels fail to reject the ADF unit-root null and reject the KPSS stationarity null - consistent with the visible trending/regime behaviour. Log returns are stationary on both tests. **Implication:** a change point model applied to price levels detects shifts in the mean price level (regime shifts), which is the quantity of direct business interest; log returns are the right series for any future volatility-regime or ARIMA/GARCH work.

4.3 Volatility

Log returns (Fig. 2) show clear **volatility clustering** - large swings bunch together in time around 1990-91, 2008-09, 2014-16, and 2020, rather than being spread uniformly - and a 30-day rolling standard deviation (Fig. 3) makes these high-volatility windows explicit. The return distribution (Fig. 4) is strongly fat-tailed (excess kurtosis ~ 66) and left-skewed (skew ~ -1.74): large negative shocks are more extreme and more frequent than large positive ones.

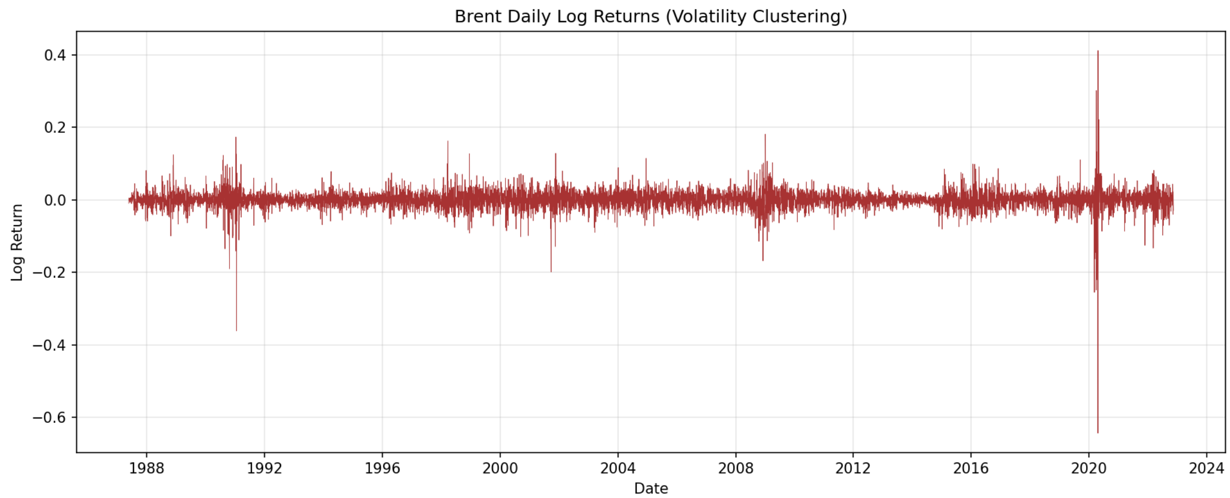

Figure 2. Daily log returns, showing volatility clustering.

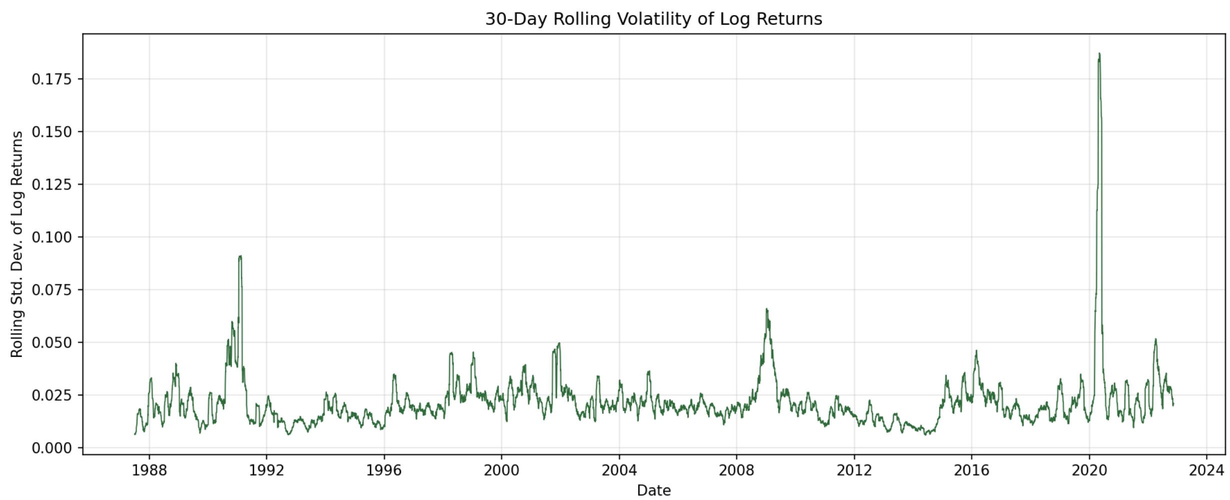

Figure 3. 30-day rolling standard deviation of log returns.

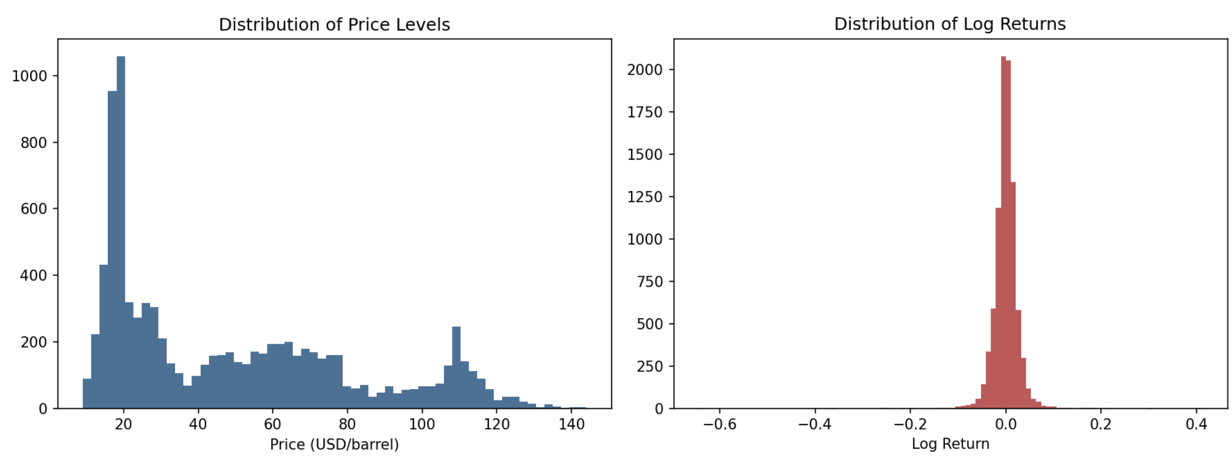

Figure 4. Distribution of price levels (left) vs. log returns (right).

4.4 Modeling Implications

- Non-stationary price levels + multiple visible regimes -> a **change point model on price levels** (not a single global mean/trend model) is the right tool to identify structural breaks.
- Fat-tailed, clustered volatility -> a Normal likelihood is a reasonable starting point for the Task 2 mean-shift model, with a Student-t likelihood and/or explicit volatility-regime (Markov-switching) modeling as a natural extension.

Full reproducible analysis: *notebooks/eda.ipynb* (executed, with outputs) and *scripts/eda.py*.

5. Assumptions and Limitations (summary)

Event dates are approximate; the model assumes a locally constant mean around each break; and - critically - a detected change point is a **statistical correlation in time**, not proof that a specific event caused the shift. See *docs/assumptions_and_limitations.md* for the full discussion, including what would be required to move from correlation to a causal claim (plausible mechanism, ruling out confounders, counterfactual/control-series reasoning).